FRE-GT-9743 Assignment 6

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1 Question.

Consider an EUR/USD FX forward collateralised in GBP. Derive its pricing framework and address the following questions:

1. What is the valuation formula?
2. What is the fair forward strike?
3. Suppose the curve construction provides only the following discount factors:

$$\mathrm{df}^{\mathrm{SONIA}}, \quad \mathrm{df}^{\mathrm{EuroSTR}}, \quad \mathrm{df}^{\mathrm{SOFR}}, \quad \mathrm{df}^{[\mathrm{GBP},\,\mathrm{USD}]}, \quad \mathrm{df}^{[\mathrm{EUR},\,\mathrm{USD}]}.$$

   Are these sufficient to compute the forward's value and fair strike under GBP collateralisation? Explain your reasoning.

Implement library contents

Github Link: https://github.com/Jerry0108-SJ/FRE-GY-9743-Assignments-6.git
Please follow the instruction in the notebook and complete the code.